Value Proposition

Adaptive Fitness Intelligence — National AI Challenge

Prepared by Manus AI · 20 August 2026

1 Value proposition

Adaptive Fitness Intelligence turns scattered wearable, nutrition and lifting records into one transparent daily training decision: what to do next, what changed, and which signals justify that recommendation.

1.1 The user-level promise

For a lifter following a defined block, the product makes the next session simpler to run and easier to trust. It retains split, movement, load, repetitions, set-level RPE/RIR, comments and the last successful comparable session. It then places that history alongside device-normalised recovery and sleep trends, nutrition adherence and the current objective—strength, hypertrophy, fat loss, endurance support or return-to-training.

1.2 The outcome

Instead of saying only "you are 62% recovered", the experience would say: **Repeat last week's top set rather than add load. Your 7-day resting-heart-rate trend is above baseline, sleep has fallen, and the prior session finished at RPE 9. Confidence: moderate.** The recommendation remains editable and explicitly non-clinical; it reduces decision friction while retaining athlete agency.

1.3 Why this is credible

Resistance-training literature supports practical, context-rich signals such as sessional and RIR-based RPE for individualisation, while warning that many physiological measures are impractical or insufficient alone. [1] A 2022 meta-analysis found no statistically significant overall 1RM advantage for autoregulated versus standardised load prescription across 15 studies. The product must therefore be framed as decision support that makes individualisation legible—not as a proven superior replacement for good programming or coaching. [2]

1.4 Minimum magical loop

Step	User value
1. Connect	Consent to Apple Health/WHOOP data or import a CSV; see clearly what is being used.
2. Log	Enter or paste a session in notes-style notation; the app structures it and surfaces ghost sets.
3. Compare	View recovery, nutrition and completed training dose together.
4. Decide	Receive a bounded next-session suggestion aligned to the active block goal.
5. Learn	Track which adjustments were followed and how performance responded.

2 References

1. Helms et al., **Methods for Regulating and Monitoring Resistance Training** (2020).
2. **The Effect of Load and Volume Autoregulation on Muscular Strength and Hypertrophy: A Systematic Review and Meta-Analysis** (2022).